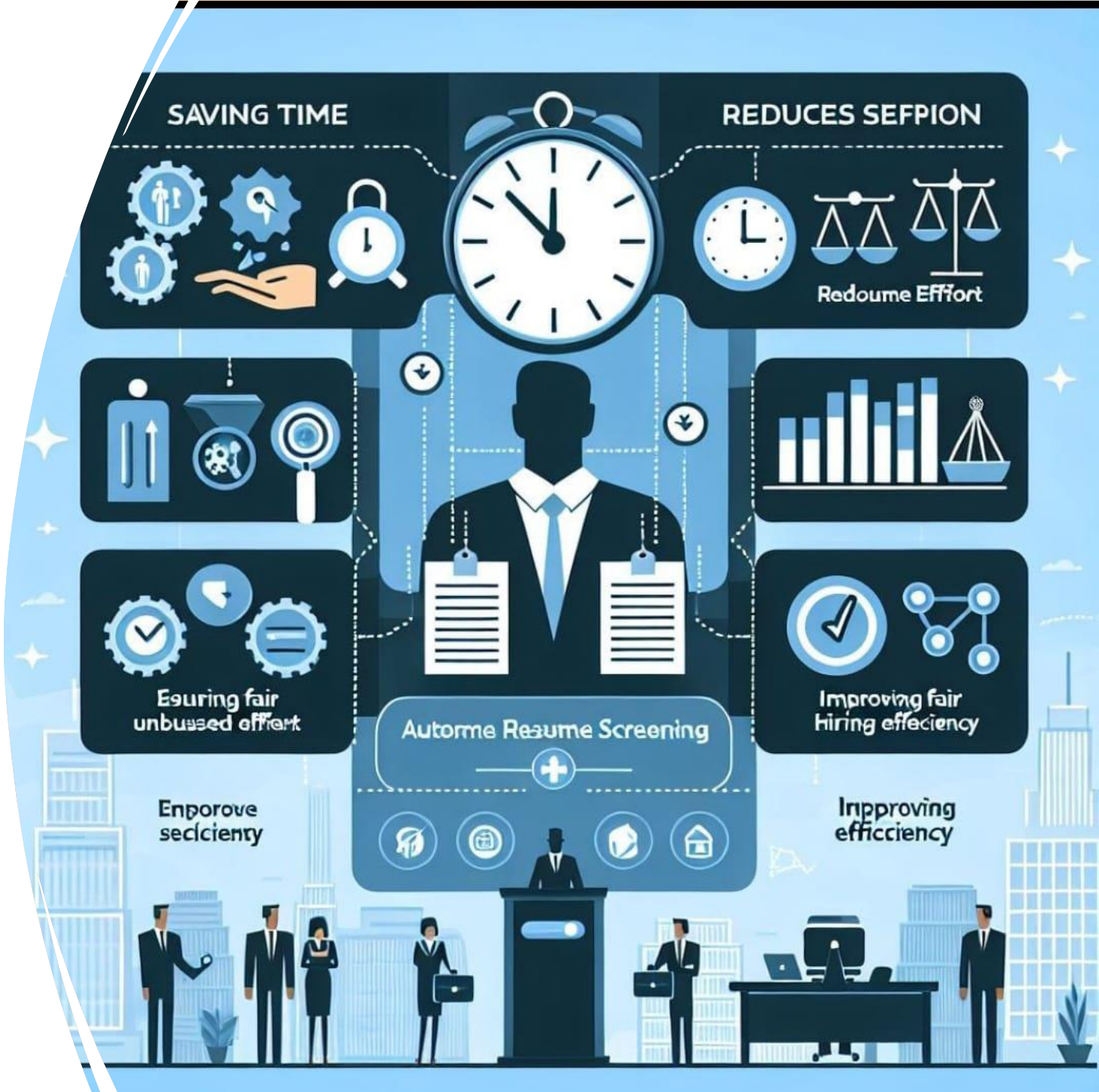
Resume Shortlisting System

using NLP, TF-IDF & Cosine Similarity

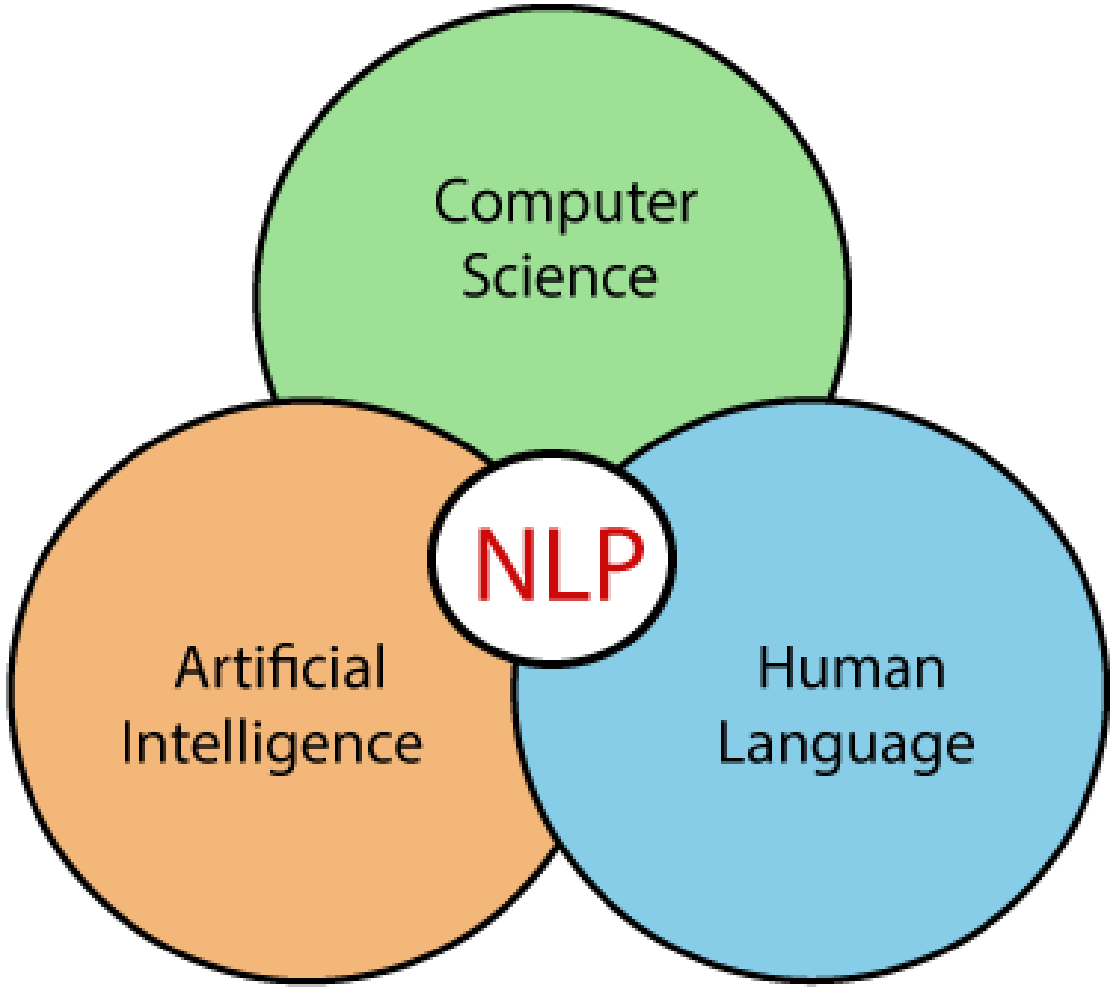


Presented by Parnika K

Introduction

- **WHAT IS NLP ?**

- **NLP = Natural Language Processing**
- It is a field of computer science that helps **computers understand human language** (like English).



Where NLP is used in Data Analysis

Text Cleaning

Remove

- symbols
- Stop words
- unnecessary text

Agenda

- The agenda of this project is to build an intelligent system that automatically analyzes and ranks resumes based on their relevance to a given job description using Natural Language Processing techniques.

Problem Statement:

- Companies get many resumes
- Manual checking is slow and difficult

Objective:

- Automates resume screening
- Finds best candidates quickly

TECHNOLGIES USED:

Python
Pandas
NLTK
Scikit-learn

Working:

Text preprocessing
TF-IDF
Cosine similarity
Ranking

- **Advantages:**
- **Saves time**
- **Reduces manual work**
- **Accurate**
-

- **Future Scope:**
- **Use AI models (BERT)**
- **Build web app**
-

Result

resume presenration.pptx | Home | Resume Automation

localhost:8888/notebooks/Resume%20Automation.ipynb

jupyter Resume Automation Last Checkpoint: 19 hours ago

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Code

JupyterLab Python 3 (ipykernel)

1	J2	Data Scientist	python machine learning statistics
2	J3	Clinical SAS Programmer	sas clinical sdtm adam tlf

[44]:
```
stop_words = set(stopwords.words('english'))

def clean_text(text):
    text = str(text).lower()   # lowercase
    text = re.sub(r'[^a-zA-Z]', ' ', text)  # remove symbols

    words = text.split()
    words = [w for w in words if w not in stop_words]

    return " ".join(words)
print(stop_words)
```

```
{'we', 'isn', 'herself', 'ma', "won't", 'above', 're', 'down', 'yourself', "i'm", 'such', 'because', "you'd", 'his', 'did', 'here', 'hers', 'at', 'how',
'those', 'its', 'has', 'needn', 'does', 've', 'these', 'hasn', 'been', 'am', 'itself', 'why', "it'll", "wasn't", 'don', "it's", 'off', "isn't", 'or', 'd
o', 'is', 'below', 'her', 'weren', 'wasn', 'all', "you've", 'me', 'o', "should've", 'they', 'themselves', 'couldn', 'be', 'again', 'i', "needn't", 'won',
'for', "i'll", 'wouldn', 'my', 'very', 'whom', "you're", 'shan', "don't", "mustn't", "they're", "hasn't", 'of', 'haven', 'she', 'myself', "he's", "it'd",
'ourselves', 'that', 'not', 'yourselves', 'where', 'by', 'to', 'being', 'can', "doesn't", 'have', 'more', "they'll", "they've", "they'd", "you'll", 'thei
r', 'both', "didn't", 'any', 'no', "shouldn't", 'only', 'before', 'now', 'through', 'some', 'shouldn', "we'd", "she'll", 'on', 'most', 'them', 'during',
'which', 'and', 'having', "she'd", 'your', 'further', 'too', "aren't", 'few', 'were', 'it', 'so', 'once', 'there', 'than', 'had', 'own', "i'd", 'same',
'mustn', 'just', 'was', "weren't", "that'll", 'aren', 'about', 'y', 'he', 'the', 'what', 'this', "mightn't", 'himself', 'nor', "we'll", 'will', 'other',
'you', 'll', 'then', 'mightn', 'into', "haven't", "shan't", 'are', 'yours', 'ours', 'who', "she's", "we've", "we're", 'an', "wouldn't", 'out', "he'll",
'ain', 'doing', "i've", 'over', 'if', 'as', 'with', 'but', 't', 'd', 'after', 's', "couldn't", 'should', 'hadn', 'up', 'under', 'a', 'theirs', 'from', 'b
etween', 'while', "hadn't", 'our', 'against', 'when', 'him', 'each', 'didn', 'until', 'm', 'in', "he'd", 'doesn'}
```

[45]:
```
data['clean_resume'] = data['resume_text'].apply(clean_text)

data[['resume_text', 'clean_resume']].head()
```

[45]:

	resume_text	clean_resume
0	deep learning tensorflow keras python	deep learning tensorflow keras python

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Conclusion

In conclusion, this project successfully demonstrates how Natural Language Processing can be used to automate the resume screening process. By using techniques like TF-IDF and cosine similarity, the system efficiently compares resumes with job descriptions and ranks candidates based on their relevance.

Any questions?

Thank You